

**SIGNSIGHT: EMPOWERING ACCESSIBILITY WITH
TAMIL SIGN LANGUAGE RECOGNITION USING
MACHINE LEARNING**

25-26J-404

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specializing in
Information Technology

Department of information technology

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

**VOICE/AUDIO TO TAMIL SIGN LANGUAGE (TSL)
CONVERSION SYSTEM**

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Declaration

I confirm that the work in this proposal is my own. It hasn't been submitted for any degree or qualification at another university or institution. As far as I'm aware, it doesn't include any material written or published by someone else, unless clearly acknowledged and cited in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:

Date: 31.08.2025

Abstract

This research is dedicated to the development of a recognition system for Tamil Sign Language (TSL), designed to convert spoken Tamil audio into visual TSL gestures using machine learning techniques. The primary goal of this work is to bridge the communication gap faced by hearing-impaired individuals by enabling them to understand spoken Tamil through real-time visual representations. The system is structured into three core modules: a speech recognition module that transcribes Tamil audio into text, logistic regression models that predict TSL gestures from the tokenized text, and a user-friendly API capable of translating multimedia inputs into TSL gestures. The approach emphasizes lightweight and scalable machine learning models, ensuring effectiveness in low-resource environments. Tamil text is tokenized and mapped into compatible TSL gesture representations, creating a seamless alignment between spoken language and sign output. Furthermore, the system supports real-time conversion, providing a practical and accessible tool for everyday use by the hearing-impaired community.

This research focuses on inclusivity, accessibility, and social impact, aiming to empower Tamil-speaking hearing-impaired persons and reduce barriers in education, healthcare, and interpersonal communication. By introducing an innovative audio-to-sign pipeline tailored for Tamil, the project contributes to advancing assistive technologies for underrepresented languages, fostering greater equality and accessibility in society.

Keywords - Tamil Sign Language (TSL), speech recognition, Tamil audio-to-text conversion, machine learning, logistic regression, tokenization, real-time conversion, assistive technology, inclusivity, accessibility, low-resource environment, hearing impaired communication, visual representation, Tamil-speaking community, multimedia input, scalable models.

TABLE OF CONTENTS

Declaration	3
Abstract.....	4
1. LIST OF FIGURES	7
2. LIST OF TABLES	7
3. LIST OF ABBREVIATIONS	8
4. INTRODUCTION	9
5. Background and Literature Survey.....	10
6. Research Gap.....	12
7. Research Problem.....	14
8. OBJECTIVES.....	15
9.1 Main Objective	15
8.1 Specific Objective.....	15
9. METHODOLOGY	16
9.1 Software Solution	17
9.1.1 Development Process.....	17
9.1.2 Feasibility Study	17
9.1.3 Requirements Gathering and Analysis.....	19
9.1.4 Data Set.....	19
9.1.5 Implementation	20
9.1.6 Testing.....	21
9.1.7 Gantt Chart.....	22
9.1.8 Work Breakdown Chart	22
10. Project Requirements.....	23
2 BUDGET AND BUDGET JUSTIFICATION.....	25

Table 7.1 Estimated budget table.....	25
• Data Collection - LKR 4,000	25
• Software Development - LKR 4,500.....	25
• Testing and Validation - LKR 4,500.....	25
• System Deployment – LKR 2,500	26
7 REFERENCES.....	27

1. LIST OF FIGURES

Figure 9.1 System Architecture.....	16
Figure 9.1.1 Component Overview Diagram	20
Figure 9.1.7 Work Breakdown Structure	15
Figure 9.1.8 Gantt Chart.....	15

2. LIST OF TABLES

Table 5.1 Description of Research Gap	12
Table 11.1 Estimated budget table.	25

3. LIST OF ABBREVIATIONS

Abbreviations	Description
ASR	Automatic Speech Recognition
TSL	Tamil Sign Language
AI	Artificial Intelligent
ITS	Intelligent Tutoring Systems
AWS	Amazon Web Services
PWA	Progressive Web App
CDN	Content Delivery Network
SUS	System Usability Scale
NLP	Natural Language Processing
CNN	Convolutional Neural Network
ML	Machine Learning

4. INTRODUCTION

Communication is the foundation of human connection. For people with hearing impairments, however, daily communication can often become a barrier, especially in societies where sign languages are underrepresented and not well supported by technology. [6] Tamil Sign Language (TSL) is one such language. It is the primary form of communication for many Tamil-speaking hearing-impaired individuals, yet it has not benefited from the same level of technological advancement as larger sign languages like American Sign Language (ASL) or British Sign Language (BSL). This lack of support makes it difficult for hearing-impaired persons to interact freely in educational settings, workplaces, healthcare services, and everyday conversations.

Imagine a Tamil-speaking parent who is deaf trying to communicate with a teacher at their child's school. Without an interpreter, the parent might rely on written notes or gestures, which often leads to misunderstandings. Similarly, think of a hospital where a deaf patient struggles to explain their symptoms to a doctor who does not know TSL. These everyday challenges show why bridging the communication gap is not just a matter of convenience it is about inclusion, equality, and dignity.

In the last decade, research worldwide has made progress in developing tools for sign language interpretation. For example, mobile apps have been created to translate text messages into sign animations, and researchers have built 3D avatars that can perform signs to improve digital communication. [1] In Sri Lanka, Sinhala Sign Language has seen attempts at text-to-sign systems and gesture recognition models. While these are important milestones, most of the existing solutions still rely on either text or visual gesture input. Very few systems attempt the more challenging task of translating spoken language directly into sign language in real time.

Tamil, like Sinhala, is a linguistically rich and phonetically complex language. It has multiple dialects and accents, and speakers often mix Tamil with English or Sinhala words in casual conversation. These factors make automatic speech recognition (ASR) a challenging task. Converting spoken Tamil into TSL adds another layer of difficulty, as the system must not only transcribe speech but also map it into meaningful TSL gestures. Unlike spoken or written Tamil, TSL does not follow a word-for-word translation. [1] For example, a spoken sentence like *"I am going to the market now"* might be expressed in TSL with just three signs *"I - Market - Go"*. This difference requires intelligent mapping that understands both language and cultural context.

The novelty of our approach lies in building a direct audio-to-sign pipeline for Tamil, rather than going through text interfaces that require reading ability or gesture-only systems that rely on cameras. By combining speech recognition with lightweight, scalable machine learning, we aim to design a solution that works in real-world environments whether it is a noisy classroom, a busy hospital, or a simple conversation at home. [2][3]

This project does not only serve as an assistive technology; it also represents a step forward for research in low-resource languages. Much of today’s AI development is concentrated on widely spoken languages like English, Mandarin, or Spanish. By focusing on Tamil, we highlight the importance of inclusivity in technology and contribute to global efforts in creating systems for underrepresented communities.

Ultimately, our goal is to empower the Tamil-speaking hearing-impaired community with a tool that makes communication easier and more natural. Just as subtitles in movies have opened doors for millions, real-time Tamil audio-to-sign conversion can open new opportunities in education, employment, healthcare, and daily life ensuring that no one is left behind simply because of the language they use to communicate.

5. Background and Literature Survey

Tamil Sign Language (TSL) technology platforms have historically concentrated on static gesture recognition and vision-based translation systems, with limited exploration of direct audio-to-sign conversion pipelines. Early TSL research by Sudha and JothiLakshmi explicitly identifies this limitation, focusing primarily on static hand gesture recognition using shape descriptors and achieving 90% accuracy with Naive Bayesian classifiers, but acknowledging the challenge of dynamic sign recognition and real-time processing [2]. Their vision-based approach represents first-generation TSL systems that require visual input rather than addressing the more complex challenge of spoken language interpretation.

Across broader assistive technology research, recent advances in low-resource language processing have demonstrated the potential for speech-to-sign conversion in underrepresented linguistic communities. A 2024 systematic review on sign language recognition technologies emphasizes that "real-time sign language detector" systems can significantly improve "communication between the deaf and the general public", while highlighting persistent challenges in continuous sign recognition and real-world deployment [11]. The review identifies that most current systems still rely on vision-based gesture capture rather than audio input, creating accessibility barriers for users who cannot perform signs or lack camera-equipped devices.

Within Tamil-specific research, contemporary work has focused primarily on recognition rather than generation systems. The development of the TLFS23 Tamil fingerspelling dataset represents significant progress, creating "a labelled dataset for Tamil Fingerspelling named TLFS23 has been created for research related to vision-based Fingerspelling translators", though it remains limited to fingerspelling recognition rather than full sign language generation [4]. Recent machine learning approaches for TSL recognition have achieved promising results using CNN architectures, but these systems continue to emphasize gesture-to-text translation rather than the inverse audio-to-sign pipeline [8].

From a linguistic complexity perspective, Tamil presents unique challenges for automatic speech recognition that directly impact audio-to-sign conversion systems. Tamil's phonetic complexity, with 247 characters and multiple dialectal variations, requires robust ASR models capable of handling code-switching between Tamil, English, and regional languages [4,7]. Additionally, TSL's non-linear grammatical structure, where spoken sentences undergo significant syntactic transformation into sign sequences, demands intelligent semantic mapping rather than direct word-for-word translation [12].

For low-resource language technologies, recent work highlights the specific challenges of developing AI systems for underrepresented communities. "Lighting low quality, lack clarity, and noisy background of the image can lead to low recognition performance" in vision-based systems, while audio-based approaches face similar environmental challenges with background noise and accent variation [11]. These findings motivate the need for robust, multi-modal approaches that can function effectively in real-world conditions.

Cross-linguistic sign language research demonstrates that successful systems must address four critical technical challenges: real-time processing latency, cultural and dialectal adaptation, environmental robustness, and scalable deployment architectures. Recent developments in South African sign language translation for COVID-19 communication emphasized the importance of "real-time speech-to-text to sign language translator" systems that can function in high-stakes communication scenarios [2].

Takeaway for TSL audio-to-sign systems. Evidence converges on four design imperatives for next-generation TSL platforms:

- **Direct Audio Processing:** Moving beyond vision-dependent systems to enable hands-free, audio-driven sign generation that accommodates users with varying physical capabilities
- **Real-time Semantic Mapping:** Implementing intelligent translation algorithms that understand TSL's unique grammatical structure and cultural context rather than literal word-for-word conversion
- **Dialectal Robustness:** Developing ASR models specifically trained on Tamil dialectal variations and code-switching patterns prevalent in everyday speech
- **Environmental Adaptability:** Creating systems that maintain accuracy across diverse acoustic environments, from quiet homes to noisy public spaces

These imperatives directly address the "audio-to-sign conversion gap" observed in current TSL research and align with broader accessibility goals for low-resource sign language communities [4,8,2].

6. Research Gap

Although sign language technologies have seen steady progress globally, the space for Tamil Sign Language (TSL) remains underdeveloped. Existing studies on sign language recognition, whether focused on gesture detection or text-to-sign translation, are either limited in scope or resource-intensive, and do not provide a real-time, speech-driven solution for Tamil.

For instance, research in gesture recognition often employs convolutional neural networks (CNNs) to classify static or predefined signs. While effective in constrained settings, these approaches do not capture the dynamic and context-dependent nature of natural TSL communication. Similarly, mobile and desktop applications for sign language interpretation such as those developed for American Sign Language (ASL) or British Sign Language (BSL) tend to rely on text-to-sign pipelines or image-based datasets, leaving out direct audio processing altogether.

Moreover, hardware-assisted solutions such as gloves, armbands, or motion sensors have demonstrated potential in gesture recognition but fall short of delivering a full pipeline from spoken Tamil → sign output. These devices often require specialized hardware, making them impractical for everyday use, particularly in schools, healthcare centers, or households where accessibility is most needed.

Another gap is the lack of resource-efficient models. Many existing systems demand high computing power and large datasets, which limits their applicability in low-resource environments. This is particularly challenging for Tamil, where digital resources and annotated sign datasets are scarce. Tools designed for other sign languages often cannot be directly transferred to TSL due to linguistic and cultural differences in grammar, syntax, and gesture representation.

This research aims to bridge these gaps by developing a scalable, lightweight, and realtime Tamil Audio-to-TSL conversion system. By employing machine learning models such as logistic regression for mapping tokenized Tamil text to TSL gestures, the proposed solution ensures resource efficiency while maintaining accuracy. Unlike existing systems that rely solely on text or gestures, this approach integrates speech recognition with gesture prediction into one end-to-end pipeline, making it both practical and inclusive for Tamil speaking communities.

Topic	What Exists Now	What's Missing (Gap)	How This Research Will Help
Sign Recognition	CNN-based gesture recognition systems classify static/predefined signs.	Cannot handle dynamic, context-rich TSL communication.	Develop a system that maps speech-driven input to flexible, real-time TSL gestures.
Speech-to-Sign Translation	Most systems (global and regional) focus on text-to-sign or image-based translation.	No system directly converts spoken Tamil into TSL in real time.	Build an end-to-end pipeline that translates Tamil audio directly into TSL gestures.

Hardware-Assisted Systems	Gloves, armbands, and motion sensors used for gesture recognition.	Require specialized hardware; no complete speech-to-sign conversion.	Deliver a software-only solution that works on standard devices without extra hardware.
Accessibility & Efficiency	Existing fingerspelling/sign systems are resource heavy and dataset dependent.	Inaccessible in low resource environments; Tamil datasets are scarce.	Use lightweight ML (logistic regression) and tokenization for efficient, low-resource operation.
Audio Integration	Image-based or gesture only recognition systems.	Do not process Tamil speech; exclude spoken language as an input.	Integrate Tamil speech recognition with gesture prediction to achieve full audio-to-sign flow.
Scalability & Usability	Tools rely on predefined datasets or offline processing (e.g., limited mobile apps).	Lack flexibility to adapt across different speakers, dialects, and real-world scenarios.	Ensure real-time, adaptive performance across diverse speech patterns and accents.

Table 6.1 Description of Research Gap

7. Research Problem

Despite the development of sign language recognition and learning technologies, the learners of TSL still experience education and communication barriers. Mobile and web based platforms like Master-ASL [1] have shown that sign lessons with quizzes can take place through mobile and web-based apps, but most of them cover the American Sign Language and offer partial support to Tamil. These systems are based on the recognition and evaluation of the static nature and hence the gap in providing the corrective feedback in real-time in dynamic and natural signing.

Studies on AI-based education indicate that the most effective features are individualized tutoring, in-person communication, and feedback in a timely manner [2][3]. These features are not commonly exercised by existing sign-learning applications, however. Students are used to training without being adjusted to their speed or mood, and there is little opportunity to achieve engagement-conscious coaching and an individual lesson planning, both important on motivation and long-term improvement.

Multi-modal translations approaches, including the Tamil speech-to-sign pipeline by Kowsigan [5], show that it is possible to perform multimodal translations, but not the pedagogy. They have difficulties with dialectal differentiation, expressiveness, and responsiveness in the moment. Traditional web learning platforms [4] are personalized and interactive yet fail to answer the needs of the TSL education. This results in an imperative research gap: the absence of a scalable, browser based TSL platform that combines real time coaching, conversational role-play, personal adaptation and explainable teacher parent insights to ensure learning is effective and inclusive.

8. OBJECTIVES

9.1 Main Objective

The main goal of this research is to create a comprehensive system that bridges the communication gap between Tamil-speaking hearing-impaired individuals and the wider hearing population. The system will consist of three main components: (1) converting spoken Tamil audio into text using automatic speech recognition (ASR); (2) mapping the transcribed and tokenized Tamil text into corresponding Tamil Sign Language (TSL) gestures using machine learning techniques such as logistic regression; and (3) providing a user-friendly API and visualization module that converts audio or video inputs into real-time TSL gestures. These components are designed to give hearing-impaired users accurate, timely, and accessible sign representations of spoken Tamil, enabling them to participate more effectively in education, healthcare, workplaces, and daily conversations.

The goal of building a machine learning model that can precisely map Tamil speech to TSL gestures is to ensure reliable real-time communication. The system will be trained using a dataset of Tamil speech samples and corresponding TSL gestures to understand the linguistic and cultural mapping between spoken Tamil and sign representations. Once trained, the system will be able to recognize speech, tokenize it, and output TSL gestures that preserve meaning while adapting to sign language grammar. This will allow hearing-impaired users to follow speech as it happens, without relying on intermediaries such as interpreters.

The ultimate objective is to offer the Tamil-speaking hearing-impaired community a dependable, scalable, and resource-efficient tool that enhances inclusivity, reduces social isolation, and ensures equal access to education, healthcare, and public services.

8.1 Specific Objective

- Build a speech recognition module to convert Tamil audio to text using machine learning algorithms
- Train logistic regression models to predict Tamil Sign Language gestures from tokenized text input
- Develop a user-friendly API for converting multimedia inputs into Tamil Sign Language gestures
- Tokenize and map Tamil text to legacy character representations for TSL compatibility
- Introduce scalable and low-resource models to map auditory inputs to visual signs
- Enable real-time conversion of audio and video content into visual TSL for improved accessibility

9. METHODOLOGY

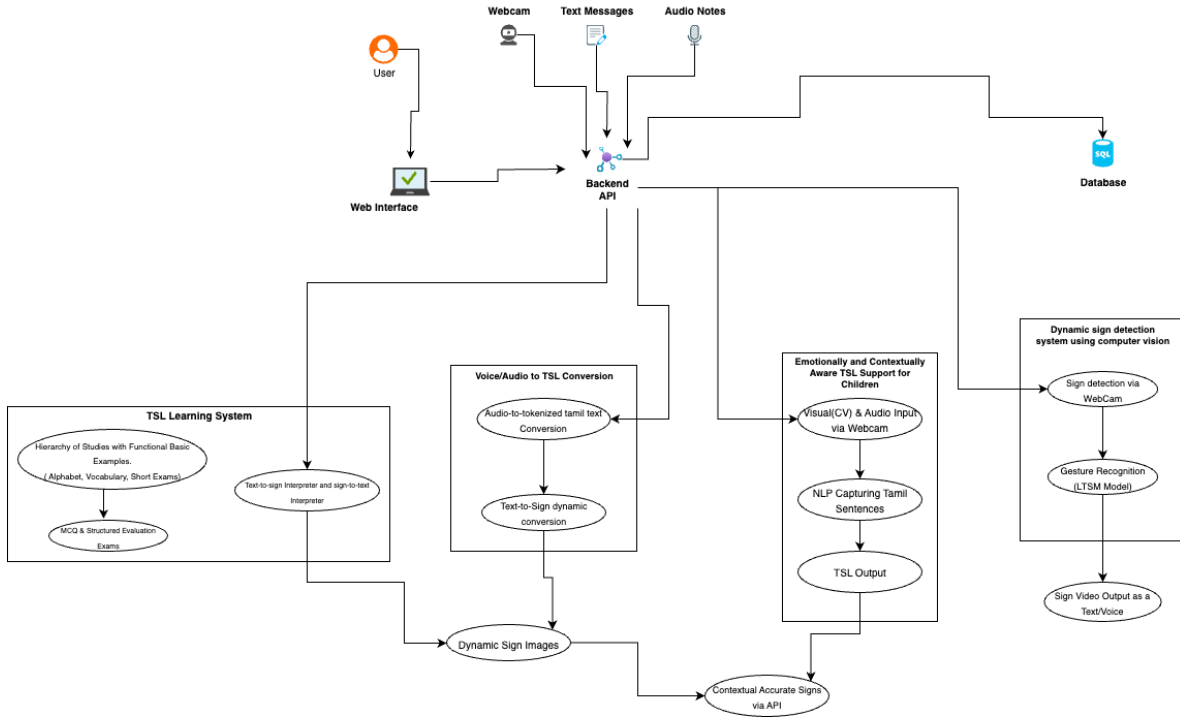


Figure 9.1 System Architecture

The development of the Tamil Audio-to-TSL Conversion Component will be implemented using the Agile methodology framework. Agile offers an iterative and adaptive approach that emphasizes continuous improvement and regular stakeholder feedback throughout the development cycle. This methodology allows the development team to build the system progressively with incremental enhancements, incorporating user feedback and addressing the specific challenges of Tamil speech processing and TSL gesture generation.

The entire development process will be organized into focused sprints, with each sprint targeting particular functionalities or components of the audio-to-TSL conversion system. Activities will include Tamil speech recognition module development, machine learning model training for audio-to-text tokenization, TSL gesture mapping algorithm creation, and backend API integration for real-time processing. Each sprint will span two weeks, followed by thorough review sessions to evaluate progress and adjust the development approach based on technical discoveries and user requirements.

The primary advantage of this approach lies in its flexibility, enabling the team to adapt methodologies based on linguistic insights, feedback from the Tamil-speaking hearing impaired community, and technical breakthroughs in Tamil speech processing and TSL gesture prediction. The development will proceed through systematic phases: Tamil audio preprocessing and noise reduction, speech-to-text conversion optimization, text-to-TSL gesture mapping using logistic regression models, real-time API integration, and comprehensive testing with diverse Tamil dialects and speaking patterns.

This structured yet adaptable methodology ensures the audio-to-TSL conversion system develops efficiently while remaining scalable and addressing the specific linguistic complexities of Tamil speech recognition and the cultural nuances of TSL gesture representation, ultimately fulfilling the communication requirements of the Tamil-speaking hearing-impaired community within the designated project timeline.

9.1 Software Solution

9.1.1 Development Process

The development of the Tamil Audio-to-TSL Conversion System will be implemented using the Agile methodology. Agile provides an iterative and adaptive framework that emphasizes continuous enhancement and regular feedback from stakeholders. This approach enables the development team to build the system incrementally with progressive refinement, incorporating user insights and addressing technical challenges as they emerge.

The complete development process will be organized into focused sprints, with each sprint addressing specific functionalities or components of the audio-to-sign conversion pipeline. Activities will encompass Tamil speech recognition module construction, machine learning model training for audio-to-text tokenization, TSL gesture mapping algorithm development, and backend API integration for real-time processing. Each sprint will span two weeks, followed by comprehensive review sessions to evaluate progress and adjust the development roadmap according to emerging requirements.

The primary advantage of this methodology lies in its adaptability, allowing the team to refine approaches based on linguistic insights, user feedback from the Tamil-speaking hearing-impaired community, and technical discoveries during Tamil speech processing and TSL gesture generation.

This structured yet flexible approach ensures the audio-to-TSL conversion system develops efficiently and remains scalable while addressing the specific linguistic complexities of Tamil speech processing and the cultural nuances of TSL gesture representation, ultimately meeting the communication needs of the Tamil-speaking hearing-impaired community within the project completion timeline.

9.1.2 Feasibility Study

9.1.2.1 Technical Feasibility

The technical feasibility of the Tamil Audio-to-TSL Conversion Component is strong, building upon well-established foundational technologies including speech recognition, machine learning, and natural language processing. The conversion of Tamil audio into text will be accomplished using advanced speech recognition APIs that support Tamil language processing, providing reliable and accurate

transcription capabilities for real-world application scenarios. The system will implement logistic regression algorithms for predicting TSL gestures from tokenized Tamil text—a proven and computationally efficient machine learning approach suitable for multi-class classification tasks in gesture recognition applications.

The system architecture is designed for optimal scalability and performance. Python serves as the primary development language for the backend API due to its robust ecosystem for machine learning and natural language processing tasks. The system will accommodate both audio and video inputs, ensuring versatility across diverse user interaction preferences. Computational scalability will be achieved through cloud service integration using platforms such as AWS or Google Cloud Platform, enabling distributed processing for machine learning model training and inference while supporting concurrent user access without performance degradation.

The technical implementation leverages Tamil-specific linguistic processing capabilities, including Unicode-to-legacy character conversion for compatibility with TSL gesture databases. The system incorporates preprocessing algorithms for noise reduction, dialect normalization, and code-switching detection to handle the complexity of real-world Tamil speech patterns. Flask framework integration ensures seamless coordination between speech recognition modules, tokenization processes, and gesture prediction models while maintaining real-time processing capabilities essential for practical deployment.

9.1.2.2 Scheduling Feasibility

The scheduling feasibility analysis demonstrates that the Tamil Audio-to-TSL Conversion Component can be developed within a realistic and manageable timeframe through systematic phase-based implementation. The project timeline is structured around three primary development phases: data acquisition and preprocessing, model development and training, and system integration with testing.

The initial phase encompasses Tamil audio dataset collection, TSL gesture image compilation, and data preprocessing operations. This phase will require approximately two months to gather sufficient training data representing diverse Tamil dialects, speaker demographics, and TSL gesture variations necessary for robust model performance. Data collection will include recording Tamil speech samples across different accent patterns and compiling corresponding TSL gesture images with appropriate labeling for supervised learning.

The subsequent phase involves machine learning model development, including Tamil speech recognition system integration, tokenization algorithm implementation, and logistic regression model training for gesture prediction. This critical development phase is allocated three to four months to ensure thorough model optimization, accuracy validation, and performance tuning. The phase includes iterative training cycles, hyperparameter optimization, and cross-validation testing to achieve reliable gesture prediction accuracy. The final phase covers system integration, API development, and comprehensive testing with user validation. This phase requires one to two months for backend integration using Flask framework, real-time processing optimization, and user acceptance testing with members of the Tamil-speaking hearing-impaired community. Additional time allocation includes performance testing across various device configurations and network conditions.

The project timeline incorporates regular milestone reviews and progress assessments to maintain development momentum. Buffer periods are strategically included to accommodate potential challenges such as data collection delays, model accuracy optimization requirements, or integration complexities. Continuous monitoring and adaptive scheduling ensure project completion within the established timeframe while maintaining quality standards and technical requirements.

9.1.3 Requirements Gathering and Analysis

9.1.3.1 Gathering requirements from Tamil schools for the hearing-impaired

The collection of requirements from Tamil-medium schools for the hearing-impaired will be a fundamental step in the development of the Tamil Audio-to-TSL Conversion System. This will ensure that the system design addresses the specific needs of the target user community. This will involve consultations with Tamil teachers, hearing-impaired students, and parents at specialized educational institutions to understand their challenges in accessing spoken Tamil content and to gather input on desired system features. Teacher contributions will inform the development of appropriate Tamil vocabulary and phrases for the conversion system. This will also assist in designing intuitive interfaces suitable for both children and adult users.

9.1.3.2 Conducting a survey

In addition to inputs from educational institutions, a comprehensive survey will be conducted among the Tamil-speaking hearing-impaired community. This will help determine the preferred input modalities that the system should support, including live audio, recorded speech, or video content. The survey will also assess community attitudes toward using an automated system for converting Tamil speech into TSL gestures and identify potential barriers to system adoption.

9.1.4 Data Set

The training of the machine learning model will utilize a dataset comprising Tamil audio recordings paired with corresponding TSL gesture images and video sequences. Audio samples must be collected across diverse speaking conditions to capture variations in pronunciation, dialects, and background noise levels. This will include labeled audio files and corresponding TSL gesture representations for each Tamil word and phrase. Each label will correspond to a specific sign language gesture sequence. The dataset will be comprehensive, with audio samples collected from multiple sources, including Tamil-speaking adults and children from specialized educational institutions, and publicly available Tamil speech datasets. The collected audio data will be preprocessed to reduce noise and normalized for consistent model training. The gesture images will be captured under controlled lighting conditions to ensure clarity. The data will be partitioned into training, validation, and test sets to evaluate model performance across different metrics. This dataset will serve as the foundation for training the machine learning model to predict TSL gestures from Tamil audio input.

9.1.5 Implementation

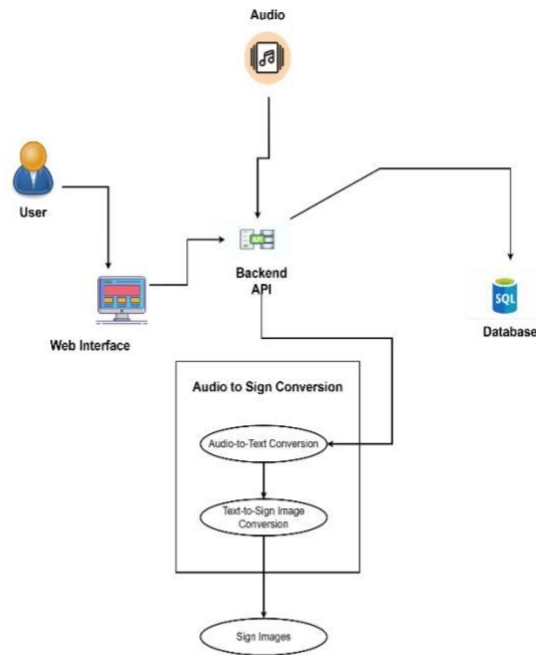


Figure 9.1.1 Component Overview Diagram

Core integrations that will be included in this phase are Tamil speech recognition, TSL gesture prediction using machine learning, and a backend API to facilitate real-time translation. The primary implementation involves developing the speech recognition module, which converts Tamil audio into structured text format. This will be accomplished using advanced speech recognition APIs that support Tamil language processing, which have demonstrated reliable performance in transcribing Tamil speech across various dialectal patterns. The preprocessing phase primarily involves tokenization, achieved by segmenting the text into meaningful units using Natural Language Processing techniques specifically adapted for Tamil linguistic structure. The system will then train a Logistic Regression model on the tokenized Tamil text to predict corresponding TSL gestures. During training, the model will be provided with labeled datasets containing TSL gesture images and their associated Tamil text inputs. This model will progressively learn to establish correlations between specific Tamil words or phrases and appropriate sign language gesture sequences.

A backend API will be developed to accept user audio input and generate TSL gesture output in real-time. This API integrates the speech recognition module that processes Tamil audio input instantaneously to deliver immediate visual feedback through TSL gesture representations. The predicted gestures are then presented through resized gesture images, providing users with clear and accessible visual representations of the translated sign language content. The API architecture ensures seamless integration between audio processing, text tokenization, and gesture prediction components while maintaining optimal response times for practical deployment scenarios.

9.1.6 Testing

9.1.6.1 Unit Testing

The system will undergo unit testing at the individual component level, including the Tamil speech recognition module, logistic regression model, and backend API. Each component will be tested independently to verify proper functionality before integration into the complete system. Unit tests will focus on validating performance at the levels of Tamil audio-to-text conversion, TSL gesture prediction accuracy, and API response reliability.

9.1.6.2 Integration Testing

Integration testing will verify that all modules operate cohesively without technical conflicts. The complete workflow from Tamil audio input through to TSL gesture output must be thoroughly tested. The system must correctly process Tamil speech input, transcribe it into tokenized text, predict corresponding TSL gestures, and display accurate visual sign language representations. Any integration-level issues will be identified and resolved during this phase.

9.1.6.3 System Testing

System testing will conduct comprehensive performance evaluations measuring system speed, accuracy, and input handling capacity. Testing will include performance assessment under realistic operating conditions, accounting for factors that may affect system performance such as Tamil dialectal variations, background noise interference, and code-switching patterns. System testing will evaluate usability aspects of the user interface, ensuring accessibility for individuals requiring assistive technologies.

9.1.6.4 User Acceptance Testing

The final testing phase will conduct User Acceptance Testing to assess system utility for the Tamil-speaking hearing-impaired community. Real users will interact with the system to evaluate accuracy, usability, and overall effectiveness in translating Tamil audio to TSL gestures. User feedback will be incorporated to make necessary adjustments, improve system performance, and enhance the overall user experience for practical deployment.

9.1.7 Gantt Chart



Figure 9.1.7 Work Breakdown Structure

9.1.8 Work Breakdown Chart

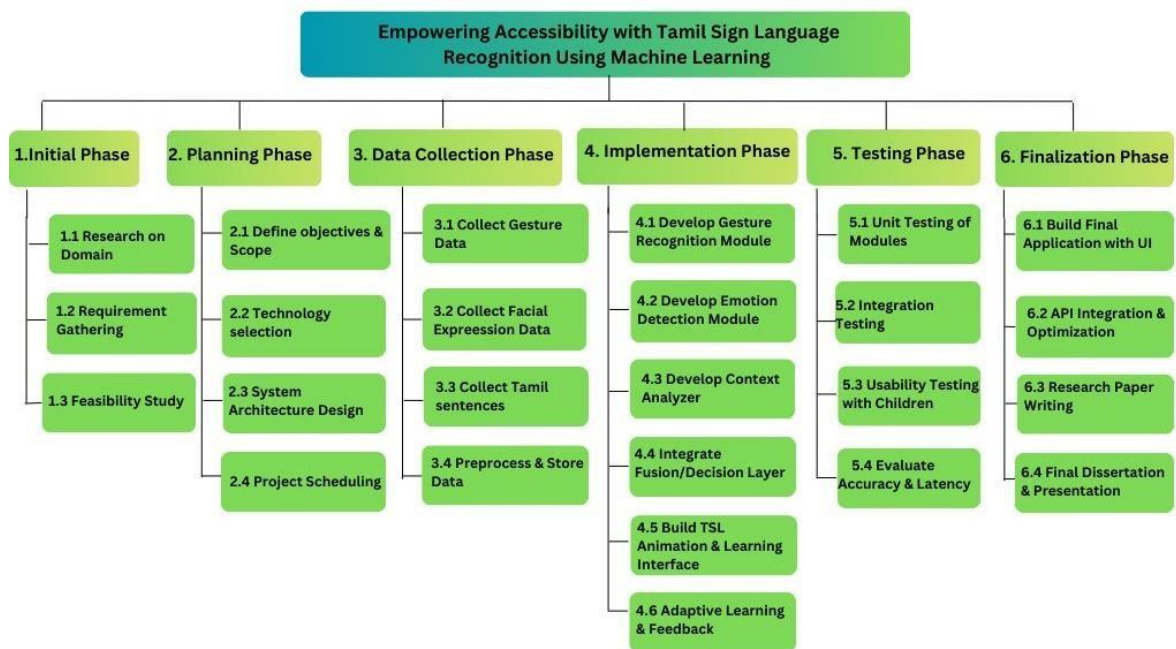


Figure 9.1.8 Gantt Chart

10. Project Requirements

10.1 User Requirements

- The user should be able to input an audio file and process the data.
- The user should be able to view the results.
- Users should be able to input Tamil speech (Voice/Audio) into the system.
- The interface should be simple, user-friendly, and accessible for deaf communities.

10.2 System Requirements

- Frontend: React.js
- Backend: Springboot
- Database: MySQL
- Development Tools: VS Code
- Speech Recognition: Google Speech Recognition API
- Libraries: NLTK (Text Tokenization), OpenCV (Image Processing)

10.3 Functional Requirements

- Convert Tamil speech to text using a speech recognition module.
- Train machine learning models to predict Tamil Sign Language (TSL) gestures from audio and text inputs.
- Develop an API to transform audio inputs into corresponding TSL gestures.
- Tokenize Tamil text and map it to legacy character representations for compatibility.
- Implement a frontend web interface using React.js for user interaction.
- Provide backend functionality using Spring Boot.
- Store and manage user data, performance logs, and audio records using MySQL.
- Process images using OpenCV for gesture recognition.
- Use logistic regression models for TSL gesture classification.
- Integrate Google Speech Recognition API for Tamil transcription.

10.4 Non-Functional Requirements

- The system should provide real-time performance for gesture recognition.
- Ensure high accuracy and efficiency in gesture predictions.
- The interface should be user-friendly and accessible.
- The backend should be scalable to support multiple users.
- The system should be secure to prevent unauthorized access to data.
- Version control and team collaboration should be maintained using Git/GitHub.
- The system should comply with accessibility standards for better usability.
- The software should be optimized for deployment on cloud or local servers.

11. BUDGET AND BUDGET JUSTIFICATION

Category	Budget (LKR)
Data Collection	4,000.00
Software Development	4,500.00
Testing and Validation	4,500.00
System Deployment	2,500.00
Total	15,500.00

Table 11.1 Estimated budget table.

The total estimated budget for the Tamil Sign Language (TSL) Audio-to-Sign Conversion System is LKR 15,500, carefully distributed across four main categories. This allocation ensures the successful collection of data, development of the system, testing, and deployment in real-world contexts.

- **Data Collection - LKR 4,000**

The system requires high-quality Tamil speech data and corresponding Tamil Sign Language gesture datasets for training and validation. Since TSL resources are limited, part of this budget will be used for curating datasets through collaboration with educators, deaf schools, and community experts. Additional costs may include transcription, annotation, and basic preprocessing of audio and gesture recordings.

- **Software Development - LKR 4,500**

The development stage includes:

- Implementing the speech recognition module (local ML-based ASR trained on Tamil datasets).
- Training logistic regression models for mapping tokenized Tamil text to TSL gestures.
- Developing a user-friendly API to connect the backend modules with the frontend interface.
- Designing the frontend application using React.js to visualize real-time gesture outputs.
- Backend integration through frameworks like Spring Boot with MySQL for database management.

This budget ensures the availability of development tools, libraries, and potential costs for external APIs during prototyping.

- **Testing and Validation - LKR 4,500**

Testing is crucial to evaluate system accuracy, performance, and usability. This includes:

- Accuracy checks on speech-to-text recognition under different accents and noisy conditions.

- Validation of gesture prediction outputs for linguistic correctness.
 - User acceptance testing with members of the Tamil-speaking hearing-impaired community.
 - Iterative improvements based on feedback from educators and accessibility experts.
- This allocation also covers logistical expenses for user trials and validation sessions.

- **System Deployment – LKR 2,500**

Deployment involves hosting the system on a secure and scalable platform. Hosting may include:

- Cloud deployment with API integration for easy accessibility.
- Providing access to schools, NGOs, and public organizations supporting hearingimpaired communities.
- Basic maintenance and monitoring of the deployed system to ensure reliability. The allocation ensures that the system is accessible to its target audience in real-world settings.

This budget allocation of LKR 15,500 strikes a balance between research rigor and resource efficiency. By prioritizing lightweight models and scalable software design, the system remains affordable while maintaining academic and practical value. The distribution emphasizes data quality, robust software development, extensive testing, and accessible deployment, ensuring that the Tamil Audio-to-TSL system is not only technically sound but also socially impactful.

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